END-TO-END BOOK ANALYTICS: WEB SCRAPING, EDA, AND PRICE PREDICTION

- **Objective:** The objective of this project is to build an end-to-end data analytics pipeline by scraping book data from an online source, performing exploratory data analysis, building a predictive model, and presenting insights through an interactive dashboard.
- **Data Collection Method:** Book data was collected using web scraping techniques from the BooksToScrape website. Python libraries such as Requests and BeautifulSoup were used to extract book attributes including title, genre, price, rating, and availability.

PHASE 1 : WEB SCRAPING (Python)

[3]:
```python
# Importing the prerequisites
import requests
from bs4 import BeautifulSoup
import pandas as pd
import time
from urllib.parse import urljoin
```

[4]:
```python
# URL Link
url = "https://books.toscrape.com/"
```

[5]:
```python
# Step-1: Fetch all genres
response = requests.get(url) # Snding HTTP request
response.encoding = "utf-8" # UTF encoding
soup = BeautifulSoup(response.text, "html.parser") # Parsing HTML using BeautifulSoup
```

[6]:
```python
# Fetching genre links
categories = soup.select("div.side_categories ul li ul li a")
```

[7]:
```python
# Creating a genre dictionary to store genre name and genre URL
genre_links = {}
for cat in categories:
    genre = cat.text.strip()
    link = urljoin(url, cat["href"])
    genre_links[genre] = link

print(f"✅ Total genres found: {len(genre_links)}")
```

```
✅ Total genres found: 50
```

[8]:
```python
# Step-2: Scrape all books genre-wise
data = []

rating_map = {
    "One": 1,
    "Two": 2,
    "Three": 3,
    "Four": 4,
    "Five": 5
}
```

```
•[11]: for genre, genre_url in genre_links.items():
           print(f"\n📖 Scraping genre: {genre}")
           page_url = genre_url

           while True:
               response = requests.get(page_url) # sending request to genre url
               response.encoding = "utf-8" # UTF encoding
               soup = BeautifulSoup(response.text, "html.parser") # Parsing HTML using BeautifulSoup

               books = soup.select("article.product_pod")

               if not books: # exit loop if no books are found on the page
                   break

               print("Books found on page:", len(books)) # print number of books found on the current page

               for book in books:
                   # Title
                   title = book.h3.a["title"]

                   # Price
                   price_text = book.select_one(".price_color").get_text()
                   price = price_text.replace("£", "").replace("Â", "").strip()

                   # Rating
                   rating_word = book.p["class"][1]
                   rating = rating_map.get(rating_word)

                   # Availability (categorical)
                   availability = book.select_one(".availability").get_text(strip=True)
```

```
                    data.append([
                        title,
                        genre,
                        float(price),
                        rating,
                        availability
                    ])

                # Pagination
                next_btn = soup.select_one("li.next a")
                if next_btn:
                    page_url = urljoin(page_url, next_btn["href"])
                    time.sleep(0.5)
                else:
                    break
```

```
📖 Scraping genre: Travel
Books found on page: 11

📖 Scraping genre: Mystery
Books found on page: 20
Books found on page: 12

📖 Scraping genre: Historical Fiction
Books found on page: 20
Books found on page: 6

📖 Scraping genre: Sequential Art
Books found on page: 20
Books found on page: 20
Books found on page: 20
Books found on page: 15
```

```
[12]: # Save to CSV
df = pd.DataFrame(
    data,
    columns=[
        "Title",
        "Genre",
        "Price",
        "Rating",
        "Availability"
    ]
)
```

```
[13]: df.to_csv("books_full_dataset.csv", index=False)

print("\n✅ Full Scrapping Completed")
print("📊 Total books scraped:", len(df))
print(df.head())
```

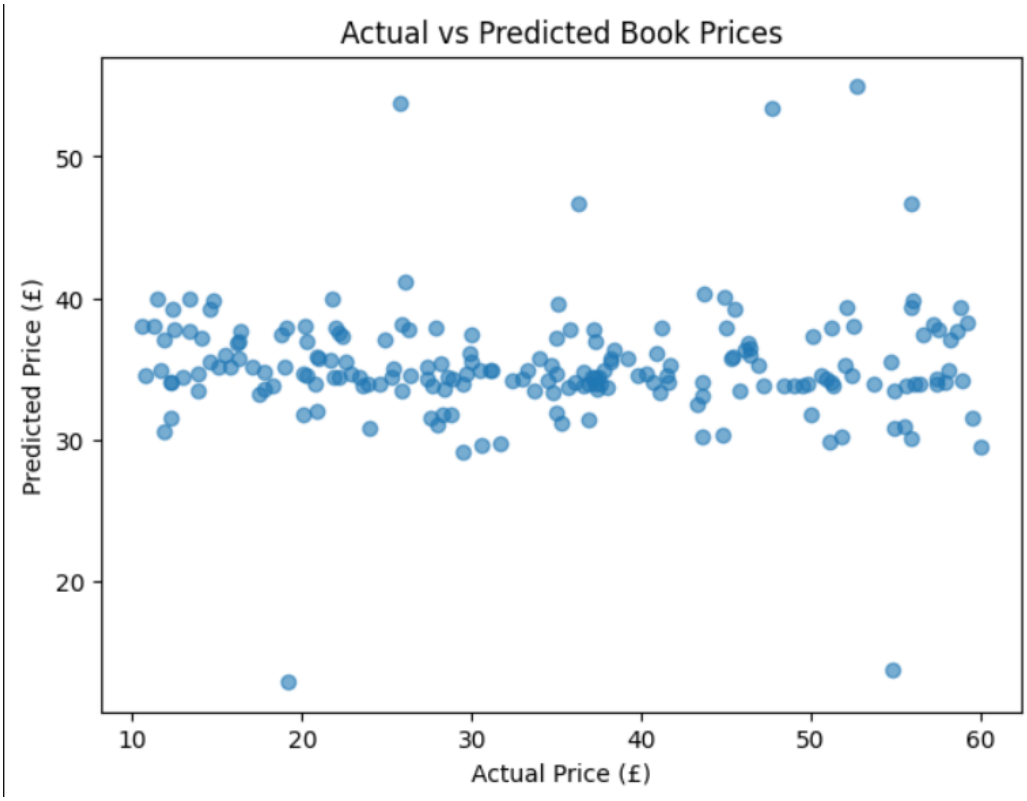
```
✅ Full Scrapping Completed
📊 Total books scraped: 1000
                                               Title   Genre  Price  Rating  \
0                            It's Only the Himalayas  Travel  45.17       2
1  Full Moon over Noah’s Ark: An Odyssey to Mount...  Travel  49.43       4
2  See America: A Celebration of Our National Par...  Travel  48.87       3
3  Vagabonding: An Uncommon Guide to the Art of L...  Travel  36.94       2
4                              Under the Tuscan Sun  Travel  37.33       3

  Availability
0     In stock
1     In stock
2     In stock
3     In stock
4     In stock
```

- **Data Cleaning and Preprocessing:** The dataset was cleaned by checking for missing values, removing duplicate records, standardizing numeric and text formats, and eliminating non-informative categories. Additional features such as price bands, rating percentages, and title length were engineered to support analysis.
- **Exploratory Data Analysis (EDA):**
  - Book prices show a **slight right-skew**, indicating the presence of premium-priced books
  - Most books are rated between **3 and 5**, suggesting generally positive feedback
  - Some genres dominate in volume but not in ratings
  - No strong relationship exists between price and rating
- **Machine Learning Model & Performance:** The regression model achieved a negative R^2 score, indicating that the selected features do not sufficiently explain price variability. This result aligns with earlier exploratory analysis, which showed weak correlations between price and other attributes. Hence, book pricing appears to be influenced by factors not captured in the dataset.

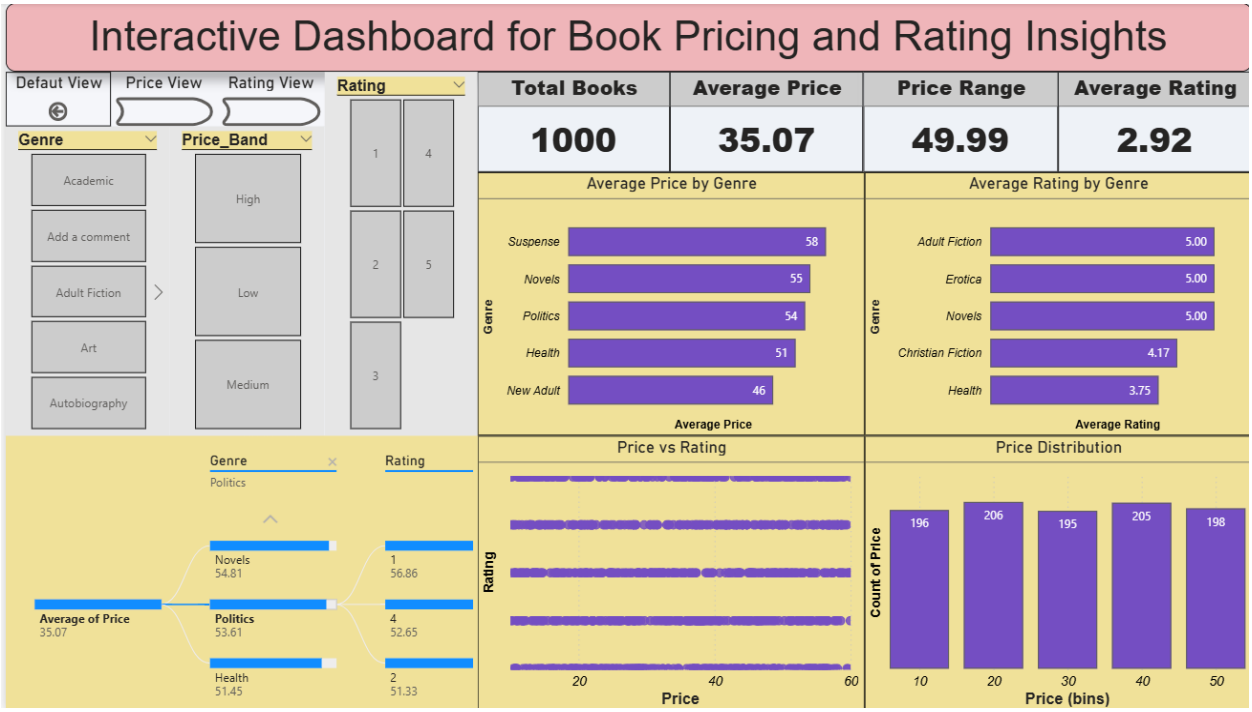
Model performance could potentially be improved by incorporating additional features such as author popularity, publisher reputation, or review text sentiment.

Metrics	Value
Mean Absolute Error	12.580858038339864
Root Mean Squared Error	15.007265817814641
R^2 Score	-0.0939837414649094



- **Interactive Dashboard:** An interactive dashboard was created using Power BI to visualize KPIs such as total books, average rating, and price range. Features including slicers, bookmarks, drill-through, and a decomposition tree were implemented to enhance user interaction and storytelling.

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- **Key Business Insights:**
  - Price does not strongly influence reader ratings
  - Genre plays a larger role in pricing than ratings
  - Premium-priced books do not guarantee higher satisfaction
- **Recommendations:**
  - Pricing strategies should be genre-specific rather than rating-driven
  - Recommendation systems should focus on content similarity instead of price
  - Future models should include author and review sentiment data

Explanation Video Link